

value of the original compound proposition. This technique is illustrated in Examples 6-8, where we also use the fact that if p and q are logically equivalent and q and r are logically equivalent, then p and r are logically equivalent (see Ex-60).

Example 6: Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

Soln: $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ (Conditional disjunction equivalence)
 $\equiv \neg(\neg p) \wedge \neg q$ (by the second De Morgan's law)
 $\equiv p \wedge \neg q$ (by the double negation law)

Example 7: Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Soln: $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ (By the 2nd De Morgan's law)
 $\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$ (By the 1st De Morgan's law)
 $\equiv \neg p \wedge (p \vee \neg q)$ (By the double negation law)
 $\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ (By the second distributive law)
 $\equiv (p \wedge \neg p) \vee (\neg p \wedge \neg q)$ (Commutative law)
 $\equiv F \vee (\neg p \wedge \neg q)$ (By the second negation law)
 $\equiv (\neg p \wedge \neg q) \vee F$ (By the 1st Commutative law)
 $\equiv (\neg p \wedge \neg q)$ (By the 2nd identity law)

Example 8: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Soln: $(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$ (Conditional-disjunction equivalence)
 $\equiv (\neg p \vee \neg q) \vee (p \vee q)$ (1st De Morgan's law)
 $\equiv \neg p \vee (\neg q \vee (p \vee q))$ (1st Associative law)
 $\equiv \neg p \vee ((\neg q \vee p) \vee q)$ (1st Commutative law)
 $\equiv \neg p \vee (p \vee (q \vee \neg q))$ (1st Associative law)
 $\equiv \neg p \vee (p \vee T)$ (1st negation law)
 $\equiv \neg p \vee T$ (1st Domination law)
 $\equiv T$ (1st domination law)